

Varun Sripad Kota
AI/ML Engineer

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SUMMARY

- AI/ML Engineer with over **4+ years** of experience, developed and deployed deep learning models using **PyTorch**, **TensorFlow**, and **Scikit-learn** for **NLP**, computer vision, and predictive analytics, enabling real-time insights and automation across complex, production-grade AI systems.
- Built cloud-native **ML** pipelines using **FastAPI**, **Docker**, **Kubernetes**, and **GitHub Actions**, integrating **CI/CD** for automated deployment, versioning, and monitoring across **AWS SageMaker** and **GCP AI Platform** environments.
- Engineered **GenAI** solutions using **GPT-4**, **BERT**, and **LangGraph**, applying **RAG** techniques, tokenization, and prompt engineering to improve contextual accuracy and response generation in enterprise-scale applications.
- Orchestrated distributed data workflows using **Apache Spark**, **Kafka**, and **Airflow**, applied **Bayesian optimization**, **A/B testing**, and **cross-validation** to tune models and drive consistent performance improvements.

TECHNICAL SKILLS

Programming Languages:	Python, SQL, Java
ML & Deep Learning:	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, LangChain, OpenCV, Natural Language Processing, Vector Databases
Generative AI & NLP:	GPT-4, BERT, LLaMA, LangGraph, RAG, Prompt Engineering, Text Generation, Tokenization
MLOps & Deployment:	MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD pipelines
Cloud Platforms:	AWS (SageMaker, Lambda, EC2), Azure (ML Studio, Databricks), GCP (AI Platform)
Data Processing & Analysis:	Pandas, NumPy, Feature Engineering, Data Cleaning, Matplotlib, Seaborn, Plotly, Statsmodels, Normalization
Big Data & Pipelines:	Apache Spark, Kafka, Airflow
Model Evaluation:	Cross-validation, Precision, Recall, F1 Score, A/B Testing, ROC-AUC
Version Control:	Git, GitHub
Other Skills:	Hyperparameter tuning, Grid search, Bayesian Optimization, Docker, Kubernetes, MLflow, Object detection, Semantic segmentation, OOP principles, Multi-threading, Parallel computing, Design patterns, Agile/Scrum methodologies

EDUCATION

Master of Science in Data Science - University of Texas at Arlington, Texas, USA
Bachelor of Science in Aerospace Engineering - Sastra University, Thanjavur, Tamil Nadu, India

WORK EXPERIENCE

Role: AI/ML Engineer | Pfizer, USA (May 2024 – Present)

- Designed scalable diagnostic pipelines using Apache Spark and Airflow to process over 100 TB of neuroimaging data. Integrated with GCP AI Platform to orchestrate multi-stage ETL and accelerate deep learning workflows. Improved throughput for cognitive research models significantly.
- Leveraged PyTorch and Hugging Face Transformers on AWS SageMaker to analyze unstructured clinical narratives from EHRs. Built custom embedding pipelines to generate longitudinal patient summaries, reducing documentation overhead and improving care plan personalization for chronic condition management.
- Developed predictive modeling frameworks using Scikit-learn and XGBoost, applying Bayesian optimization and cross-validation techniques. Tuned features and parameters to improve neurological disorder detection, increasing F1-score and diagnostic precision across multi-hospital pilot implementations and anonymized patient cohorts.
- Built interactive triage agents using GPT-4, LangGraph, FastAPI, and vector databases. Engineered retrieval-augmented generation pipelines and prompt tuning strategies, enabling real-time patient responses, and reducing clinical inquiry resolution times by 34% during care navigation trials.
- Deployed production-ready ML models using Docker, MLflow, and GitHub Actions with CI/CD integration. Enabled automated deployment of inference services across cloud-native systems with rollback support, ensuring model version control, traceability, and uninterrupted availability in regulated environments.
- Created clinician-facing dashboards using Plotly, SQL, and Statsmodels, translating complex model outputs into transparent visual summaries. Enhanced interpretability of risk scores and reduced onboarding time by enabling data-driven decision support through user-friendly, explainable AI interfaces.

Role: AI/ML Engineer | Capgemini, India (Aug 2020 – May 2023)

- Designed distributed data pipelines using Apache Spark, Kafka, and Airflow to process over 80 million records daily, enabling real-time feature generation and continuous ingestion of transactional data across multiple banking platforms and fraud monitoring systems.
- Developed entity-aware NLP workflows using BERT, Hugging Face Transformers, and custom tokenizers to extract relevant fields from KYC documents and flagged communications, improving automated document review and regulatory compliance across onboarding and customer servicing channels.
- Engineered deep learning models using PyTorch to detect anomalous financial behavior in real time, reducing false positives by 31% and supporting fraud risk assessments across dynamic datasets and evolving transaction behavior patterns.
- Trained and tuned rare-event classifiers with Scikit-learn, XGBoost, and SMOTE to handle imbalanced datasets, applying financial KPI-driven feature engineering and cross-validation to boost detection rates for fraudulent transactions and operational risk events.
- Containerized inference pipelines using Docker and deployed them with Kubernetes on AWS SageMaker, optimizing latency by 38% and supporting scalable A/B testing strategies and multi-environment deployments across global delivery ecosystems.
- Automated model lifecycle processes using MLflow, FastAPI, and GitHub Actions, delivering version-controlled scoring APIs with sub-200ms latency and reducing drift-related incidents by 25% across model retraining and production rollouts.
- Built explainable AI dashboards using Plotly, Seaborn, and SHAP to visualize prediction behavior for compliance teams, cutting audit investigation time by 19% and improving model interpretability across stakeholders and regulatory audits.
- Integrated Java-based OpenCV modules for document image parsing and parallel computing workflows, leveraging OOP principles, Agile methodology, and design patterns to enhance the robustness of multimodal risk intelligence applications.

PROJECTS

LLM-Powered Document Summarizer | Python, Hugging Face Transformers, FastAPI, Docker, AWS Lambda, Tokenization, Prompt Engineering

- Extracted and cleaned text from PDFs and HTML articles using Pandas, pdfminer, and BeautifulSoup, ensuring well-structured input for model inference by handling inconsistent formats, noisy elements, and encoding issues across diverse documents.
- Used Hugging Face Transformers fine-tuned for summarization, combining chunk-wise tokenization and prompt engineering to capture key insights from long-form documents and improve summary coherence, contextual retention, and domain-specific focus.
- Packaged the model pipeline with Docker and served endpoints through FastAPI, allowing clean integration into production workflows while maintaining modularity, stateless processing, and compatibility across different environments.
- Deployed the summarization API as a serverless application on AWS Lambda, designing retry-safe execution logic and leveraging automatic scaling to deliver summaries with sub-2-second latency under real-time, concurrent user load.
- Tested response performance using logging, concurrency simulation, and AWS CloudWatch, tuning function memory and timeout settings to maintain consistency in latency across variable-length documents and multi-user access scenarios.
- Built the system with extensibility in mind, allowing teams to plug in different LLMs or models via configuration updates, supporting enterprise use cases in legal, academic, and healthcare domains with minimal architectural changes.

Multimodal Search Engine with CLIP | Python, PyTorch, OpenAI CLIP, FAISS, Flask, Vector Databases, Semantic Search

- Curated and normalized a domain-specific image dataset, applying pre-fetch and caching logic to manage memory efficiently during large batch processing, enabling smooth offline testing and high-throughput visual retrieval performance.
- Used OpenAI CLIP with PyTorch to embed text and image data into a shared latent space, enabling zero-shot multimodal search by aligning visual and language semantics across contextually diverse queries.
- Implemented high-speed similarity search using FAISS with cosine distance metrics and GPU-based indexing, achieving top-k retrieval accuracy with consistent latency for both short and complex semantic queries across multilingual content.
- Developed a dynamic web interface using Flask, exposing RESTful endpoints for text and image queries and integrating real-time feedback with HTML templates and JavaScript for interactive testing, demos, and user-centric prototyping.
- Extended tokenization logic to handle context-rich multilingual phrases, improving intent recognition and semantic precision for search inputs through transformer-driven text encoding, enabling more accurate cross-lingual information retrieval.
- Modularized the CLIP encoder and FAISS search pipeline into reusable components, supporting future integrations with LangChain and vector databases for scalable deployment across enterprise-grade retrieval systems.